Arkabrata Roy

Software Engineer — Backend Systems — Applied AI Engineering

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Professional Summary

Final-year Computer Science undergraduate at **KIIT University** focused on **Software Engineering**, with hands-on experience in **Backend Systems and Applied AI**. Skilled in building scalable applications, developing reliable backend services, and integrating AI technologies to solve real-world problems..

Education

KIIT University, Bhubaneswar Aug 2023 – Present
Bachelor of Technology in Computer Science and Engineering **CGPA: 7.77/10**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks

Technical Skills

Languages: Python, Java, C, SQL

Backend: Node.js, Express.js, Flask, REST APIs, Microservices, gRPC, Apache Kafka

Databases: PostgreSQL, MongoDB, Drizzle ORM, Schema Design, Database Migrations

AI/ML: Generative AI, LLMs, RAG, LangChain, Hugging Face, Scikit-learn, TF-IDF, Semantic Search, Prompt Engineering

Web/Cloud: React.js, Next.js, HTML, CSS, Tailwind CSS, AWS, Docker, CI/CD

Engineering: Distributed Systems, Event-Driven Architecture, API Design, Testing, Debugging, Git, GitHub

Experience

Sundew Solutions — Full Stack Developer Intern May 2026 – July 2026
Client Project: Asmbel — U.S.-based B2B Home Warranty Platform

- **Analyzed and debugged data flows across 12 distributed microservices**, resolving service-integration issues across policy, claims, and technician-management workflows.
- **Developed and integrated REST and gRPC APIs** to connect backend services and implement cross-service business workflows.
- **Implemented asynchronous Kafka event flows**, tracing producer-consumer interactions to diagnose failures in distributed workflows.
- **Managed PostgreSQL schemas and migrations** using Drizzle ORM, coordinating database changes with dependent backend services.
- Progressed from **service-level debugging to end-to-end feature ownership**, contributing to implementation, testing, code reviews, and production-oriented fixes.

Product Manager Accelerator (Cohort 6) — AI Intern Oct 2025 – Dec 2025

- **Collaborated with a 7-member cross-functional team** to develop and validate an AI-powered MVP integrating external APIs with application workflows.
- **Converted product requirements into application features** and iterated through testing, debugging, and user feedback to refine the MVP.

Projects

BrightPath — AI-Powered School Discovery Platform Nov 2025 – Dec 2025
Next.js, LangChain, OpenAI GPT-4o-mini, OpenSearch, AWS, CI/CD

- **Built a RAG-based semantic search and recommendation pipeline** covering **100+ schools**, combining structured school data with natural-language retrieval.
- **Automated evaluation for hallucination, relevance, and response quality**, reducing manual testing effort by **60%**.
- **Deployed the application on AWS with CI/CD automation**, creating a repeatable build, test, and release workflow.

CohabitAI — Roommate and PG Allocation Platform Aug 2025 – Oct 2025
Python, Flask, Scikit-learn, Google Gemini API, MongoDB

- **Engineered a multi-factor recommendation pipeline** using Cosine Similarity, Euclidean Distance, TF-IDF, and Haversine Distance, improving matching accuracy by **28%** across **500+ test profiles**.
- **Combined constraint filtering, ranking, and KNN retrieval** to match location, preference, and compatibility requirements.
- **Integrated Gemini-generated explanations** to provide interpretable and personalized recommendation results.

Achievements and Certifications

- **Winner:** MLSA Hackathon – AI/ML Track; KPIT NOVA – Embedded Challenge.
- **State Finalist:** Smart India Hackathon 2024 — **Top 10:** ICDCIT Hackathon – NLP Track.
- **Award:** KIIT FEST – Hack The Bot.
- **Certifications:** Prompt Engineering for ChatGPT – Vanderbilt University; Generative AI: Introduction and Applications – IBM; Google Cloud Fundamentals: Core Infrastructure; Developing a Google SRE Culture.

Positions of Responsibility

Technical Secretary and Core ML Member, AISOC KIIT May 2024 – Aug 2026

- **Led GenAI workshops for 200+ students** and coordinated an AI/ML hackathon involving **50+ teams**, managing technical planning and execution.

UI/UX and ML Contributor, Elabs KIIT Mar 2024 – Aug 2026

- **Redesigned internal platform interfaces**, increasing usability by **25%**, while contributing to ML personalization modules.

Tech Lead, Project Wing, IEEE Computer Society KIIT Jan 2026 – Jun 2026

- **Led 3+ project teams** and mentored members on ML system design, deployment, and production-readiness practices.

Design Intern, KIIT E-CELL Jan 2024 – June 2024

- **Designed creative assets for 2+ events** and supported 50+ members with UX and design guidance.